Linux Plumbers Conference 2022

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HIGHMEM API and future

Architecture uConf

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HIGHMEM review

- 32bit kernels
- Large amounts of memory
- Kernel maps pages in ‘highmem’ as needed
- User space can use additional memory
  - Especially with PAE

Protection Keys Supervisor

- Overlays additional protections on kernel mappings.
- Thread/CPU local protections
- No TLB flushes
- PMEM Stray write protection abused kmap interface
  - Presents issues for long term transient direct map uses

Issues

- page_address() can’t work for PKS
- kmap()/kmap_atomic() are deprecated
  - See kmap_local_page()
- kmap_* vs page_address are confusing
- PKS is not the first user to want this

kmap_local_page()

- Can be used in any context
- Disables migration but not preemption
- Maps can only be used in the local context
- Always suitable to replace kmap_atomic()
  - kmap() calls need to be evaluated

Do we even need HIGHMEM?

- Great LWN article by Arnd Bergmann
  - https://lwn.net/Articles/838807/
- What is the need for large memory on the remaining 32bit Archs?
- Are HIGHMEM systems large core counts?

How much more should HIGHMEM API change?

- The use of page_address() will be an issue for new direct map requirements
- API Conversion on going
  - https://tinyurl.com/yc4u9vtf

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Status of HIGHMEM work

- Many call sites converted by Fabio M. De Francesco
- Always looking for more help
  - Email me or Fabio to coordinate
    - “Ira Weiny” <ira.weiny@intel.com>
    - “Fabio M. De Francesco" <fmdefrancesco@gmail.com>

Additional complications

- flush_*cache_*()
- How do we test with more distros dropping 32 bit support?
- Folios

Future

- Stop adding HIGHMEM support to new archs
- New interface to the direct map?
  - Preserve/redefine kmap?
  - Lightweight vmap?
- Alternatives to the direct map?
  - Stop mapping all of memory?
  - Performance?

Thank you

- Thank you to all the reviewers of conversions so far
- Questions, comments, and conversions welcome

References

- End to HIGHMEM – https://lwn.net/Articles/813201/
- Future of 32bit Linux – https://lwn.net/Articles/838807/
- Kmap replacement tracking sheet – https://docs.google.com/spreadsheets/d/1i_ckZ10p90bH_CkxD2bYNi05S2Qz84E2OFPv8zq__0w/edit#gid=1679714357
- Memory size is going down – https://lwn.net/ml/linux-kernel/CAK8P3a3pzgVvwyDhHPoiSOqyv+h_ixbsdWMqG3sELenRJqFuew@mail.gmail.com/

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- ARM LPAE – https://elinux.org/images/6/6a/Elce11_marinas.pdf
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